Examining the Relationship Between Antarctic Penguin Species and Body Measurements*

A comparison between Univariate ANOVA and Multivariate MANOVA

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Abstract here.

*Project repository available at: https://github.com/peteragao/MATH261A-project-template.

Introduction

"Do different penguin species differ in their overall physical characteristics, and does MANOVA provide additional insight compared with conducting separate univariate ANOVAs?"

Data

The data was collected for research that was conducted as part of the Palmer Station, Antarctica, Long-Term Ecological Research program which was supported by grants through the National Science Foundation, Office of Polar Programs (NSF-OPP). Data was originally listed as three separate datasets within the Palmer Station Long-Term Ecological Research (PAL-LTER) data system and was combined into the single Palmer penguins dataset. Data were collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER, a member of the Long Term Ecological Research Network. The data used in this analysis is provided by Horst et al. (2020). All analysis was performed in R Core Team (2025).

First we observe the dataset in Table 1. At a glance, we can see that there are some similarities between the species. Bill depth, flipper length, and body mass of Adelie and Chinstrap penguins are more similar with each other than with Gentoo. However, Chinstrap and Gentoo penguins share a similar mean bill length. Additionally, note the smaller sample size of Chinstrap penguins.

Table 1: Summary Statistics of Key Variable Means and Sample Size by Species

Table 1: Summary Statistics of Penguin Measurements

species	n	mean_bill_length	mean_bill_depth	mean_flipper_length	mean_body_mass
Adelie	151	38.791	18.346	189.954	3700.662
Chinstrap	68	48.834	18.421	195.824	3733.088
Gentoo	123	47.505	14.982	217.187	5076.016

The data is further explored in Figure 1 which shows the distribution of observations in the dataset.

Since MANOVA does not work well with strongly positively correlated variables, I calculate the correlation matrix and present the results in Table 2 From the results, we see that bill length and bill depth are weakly negatively correlated and that flipper length and body mass are strongly correlated. Since MANOVA we may expect results to be inflated

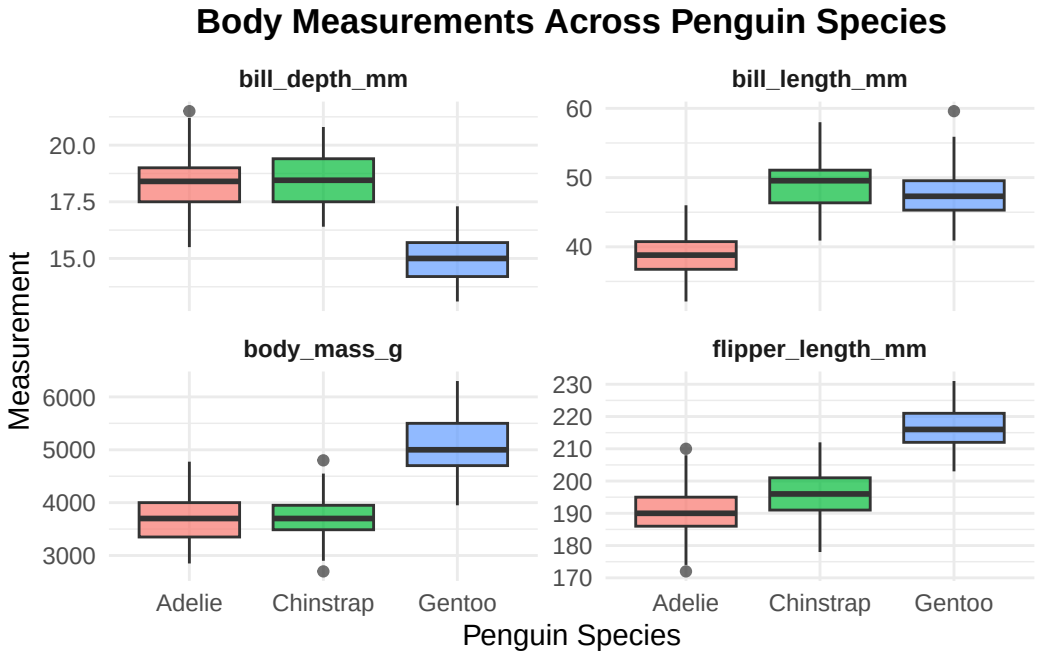


Figure 1: Visualization of Key Variable Means and Sample Size by Species

Table 2: Correlation Matrix of Penguin Measurements

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
bill_length_mm	1.000	-0.235	0.656	0.595
bill_depth_mm	-0.235	1.000	-0.584	-0.472
flipper_length_mm	0.656	-0.584	1.000	0.871
body_mass_g	0.595	-0.472	0.871	1.000

Methods

In order to investigate whether penguin species differ in their physical measurements I will consider four characteristics simultaneously: bill length, bill depth, flipper length, and body mass. First, I will start with four multiple univariate ANOVAs tested on each of the four characteristics. The four one-way ANOVAs will test whether penguin species differ across bill length, bill depth, flipper length, and body mass simultaneously. Each ANOVA tests whether the mean of one dependent variable differs across the three species groups. The general one-way

ANOVA model is

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

where Y_{ij} is the observed measurement for the j-th penguin for each species, i. μ is the overall population mean. α_i is the effect of the i-th species. ϵ_{ij} is the random error associated with each observation. This error tern is assumed to be independently and normally distributed with mean zero and variance $\epsilon_{ij} \sim N(0, \sigma^2)$. For ANOVA, we assume independence of observations, normality of residuals, and homogeneous population variance.

For each of the individual univariate ANOVAs, I will have a set of hypotheses for a total of four. My Null Hypothesis for each characteristic will be

$$H_0 : \mu_{Adelie} = \mu_{Chinstrap} = \mu_{Gentoo}$$

with an alternative hypothesis of

$$H_A : At\ least\ one\ species\ mean\ differs$$

The limitation is that each ANOVA examines one outcome independently. This means we aren't directly testing whether penguin species affects the overall combination of physical characteristics.

The general one-way MANOVA model is

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

where Y_{ij} is the observed measurement for the j-th penguin for each species, i. μ is the overall population mean. α_i is the effect of the i-th species. ϵ_{ij} is the random error associated with each observation. This error tern is assumed to be independently and normally distributed with mean zero and variance $\epsilon_{ij} \sim N(0, \sigma^2)$.

In addition to the three separate models, I will use MANOVA which considers all four dependent variables simultaneously.

MANOVA tests whether the three cylinder groups have significantly different multivariate mean vectors.

Since MANOVA evaluates the combined mean vectors, we have the null hypothesis which states that all three species have equal population mean vectors

$$H_0 : \mu_{Adelie} = \mu_{Adelie} = \mu_{Adelie}$$

and the alternative hypothesis

$$H_A : At\ least\ one\ mean\ vector\ differs$$

If the p-value associated with Pillai's Trace is below 0.05, you reject the null hypothesis.

This indicates that the penguin species differ significantly in their combined physical characteristics.

For MANOVA, we assume independence of observations, a common covariance matrix, and that each population is multivariate normal.

However, it does not identify which specific measurements differ.

MANOVA has several advantages:

It accounts for correlations between dependent variables.

It provides an overall multivariate test.

It can identify group differences that might not be apparent when looking at each outcome separately.

It avoids treating multiple related outcomes as entirely independent tests.

One particularly important point for your project: conducting three separate ANOVAs increases the overall Type I error rate unless you adjust for multiple comparisons.

MANOVA provides an overall test before you investigate individual outcomes.

Results

Table 3: One-Way ANOVA Results by Penguin Species

Dependent Variable	Degrees of Freedom	F-statistic	P-value	Significance
Bill Length	2	410.600	<2e-16	***
Bill Depth	2	359.789	<2e-16	***
Flipper Length	2	594.802	<2e-16	***
Body Mass	2	343.626	<2e-16	***

Table 4: MANOVA Results Using Pillai's Trace

Test	Pillai's Trace	Approx. F	Numerator df	Denominator df	P-value	Significance
Pillai's Trace	1.637	379.487	8	674	<2e-16	

Table 5: Univariate Follow-up Tests from MANOVA

Dependent Variable	Degrees of Freedom	F-statistic	P-value	Significance
Response bill_length_mm	2	410.600	<2e-16	***
Response bill_depth_mm	2	359.789	<2e-16	***
Response flipper_length_mm	2	594.802	<2e-16	***
Response body_mass_g	2	343.626	<2e-16	***

Discussion

You should therefore discuss the assumptions of MANOVA:

Multivariate normality within groups.

Homogeneity of covariance matrices.

Independence of observations.

Absence of problematic multicollinearity.

You should also check the correlations among MPG, horsepower, and quarter-mile time.

If your dependent variables are highly correlated, this gives you an opportunity to explain why examining their relationships simultaneously is useful.

Also, be careful not to claim that MANOVA is automatically better. Its value depends on the research question, the relationship between the dependent variables, and whether its assumptions are reasonably satisfied.

This case study demonstrates the advantages of multivariate data analysis by comparing separate univariate ANOVAs with MANOVA when examining physical differences among penguin species. While individual ANOVAs provide information about differences in specific physical measurements, MANOVA evaluates whether species differ across multiple characteristics simultaneously while accounting for relationships among the dependent variables. This provides a more comprehensive understanding of group differences and reduces the need to rely exclusively on multiple independent hypothesis tests. Therefore, multivariate analysis is particularly valuable when research questions involve several related outcome variables.

References

Horst, Allison Marie, Alison Presmanes Hill, and Kristen B Gorman. 2020. *Palmerpenguins: Palmer Archipelago (Antarctica) Penguin Data*. https://doi.org/10.5281/zenodo.3960218.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.